

GearDesign Report

Cylindrical gear(pair)

작성일: 2025년 10월 23일

Contents

1. Summary

- 1.1. Gear geometry
- 1.2. Total safety factor
- 1.3. Total damage & life
- 1.4. Others
- 1.5. Summary - Load cases

2. Detail geometry

- 2.1. Input gear geometry
- 2.2. Calculated geometry - Gear
- 2.3. Calculated geometry - Tooth
- 2.4. Calculated geometry - Mesh
- 2.5. Calculated geometry - Accuracy
- 2.6. Calculated geometry - Measurement

3. Detail rating

- 3.1. General
- 3.2. Pitting
- 3.3. Bending
- 3.4. Scuffing
- 3.5. Micro-pitting

4. Power loss

- 4.1. Mesh loss
- 4.2. Churning loss
- 4.3. Windage loss

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	20.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	20.0000	44.0000
Facewidth (mm)	b	46.0000	46.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	85.1340	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	53.2090	117.0600
Tip diameter (mm)	d_a	58.209 (58.209 / 58.109)	122.060 (122.060 / 121.960)
Root diameter (mm)	d_f	46.959 (46.767 / 46.657)	110.810 (110.617 / 110.507)
Number of teeth spanned	k	3.0000	6.0000
Base tangent length (no backlash) (mm)	W_k	19.287 (19.221 / 19.183)	42.431 (42.365 / 42.328)
Effective Diameter of ball/pin (mm)	D_M	4.5000	4.2500
Diametral measurement over two balls (mm)	M_{dk}	59.826 (59.668 / 59.577)	122.879 (122.701 / 122.598)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.1755		1.2043		1.0000
Bending S.F	3.0304		3.2350		1.4000
Scuffing S.F (Flash Temp.)	2.2033				2.0000
Micropitting S.F	0.6755				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.5000	0.0000	0.2000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4579
Overlap Ratio	2.0032
Total Contact Ratio	3.4611
Worst Pitch Line Velocity [m/s]	27.8601
Total Efficiency [%]	98.6760
Worst Mesh Loss [W]	821.8890
Worst Churning Loss [W]	1164.0872
Worst Peak-to-Peak T.E. [μm]	0.2196
Gear Total Mass [kg]	1.7878
Hunting tooth	Patial

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperautre [°C]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTe (Left) [μm]	PPTe (Right) [μm]
LC	Gear1	10000.0000	80.0000	150.0000	10000.0000	143.2390	1.1755	3.0304	2.2033				1.3000		0.8219		0.1593	0.2196	
1	Gear2			-150.0000	-4545.4550	315.1270	1.2043	3.2350									1.0048		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	20.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	20.0000	44.0000
Facewidth (mm)	b	46.0000	46.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	85.1342	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9600	0.9600
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.2000	
Transverse module (mm)	m_t	2.6604	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	21.1728	
Working transverse pressure angle (deg)	α_{wt}	21.1728	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	21.1919 / 21.1537	
Base helix angle (deg)	β_b	18.7472	
Helix angle at operating pitch circle (deg)	β_w	20.0000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Circumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0745 / 0.1171	0.0745 / 0.1171
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1405 / 0.2426	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1240 / 0.2143	
Angular backlash, min (deg)	j_θ, min	0.3025	0.1375
Angular backlash, max (deg)	j_θ, max	0.5226	0.2375
Effective facewidth (mm)	b_{eff}	46.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0385 / -0.0604	-0.0385 / -0.0604
Prevent undercut profile shift coefficient	x_{Emin}	-0.3883	-2.0542
Max. profile shift coeff. (min top land)	x_{Emax}	1.1163	1.9311
Reference center distance, $x_1=x_2=0$ (mm)	a_d	85.1342	
Working center distance (mm)	a_w	85.1342	
Reference diameter (mm)	d	53.2089	117.0596
Operating pitch diameter (mm)	d_w	53.2089	117.0596
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	53.2158 / 53.2020	117.0747 / 117.0444
27		58.2089	122.0596
Tip diameter with tolerance (mm)	$d_a, e/i$	58.2089 / 58.1089	122.0596 / 121.9596

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diameter (mm)	d _f	46.9589	110.8096
Root diameter with tolerance (mm)	d _f , e/i	46.7666 / 46.6567	110.6172 / 110.5073
Tip form diameter (mm)	d _{Fa}	58.0089	121.8596
Tip form diameter with tolerance (mm)	d _{Fa} , e/i	57.7904 / 57.6660	121.6536 / 121.5362
Root form diameter (mm)	d _{Ff}	49.9073	112.8008
Root form diameter with tolerance (mm)	d _{Ff} , e/i	49.8528 / 49.8242	112.6677 / 112.5927
Active tip diameter (EAP) (mm)	d _{Na}	58.0089	121.8596
Active tip diameter with tolerance (mm)	d _{Na} , e/i	57.7904 / 57.6660	121.6536 / 121.5362
Active root diameter (SAP) (mm)	d _{Nf}	50.1552	113.5964
Active root diameter with tolerance (mm)	d _{Nf} , e/i	50.2771 / 50.2158	113.8001 / 113.6974
Mean diameter for mass calculation (mm)	d _m	52.4353	116.2859
Gear mass (kg)	m	0.4810	1.3067
Gear inertia (kg.m ²)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.6250	0.6250
Effective tip clearance, e/i (mm)	c, e/i	0.8371 / 0.7102	0.8371 / 0.7102
Min. margin of involute contact, (d _{Nf_i} - d _{Ff_e}) / 2 (mm)	c _{Nff}	0.1815	0.5149

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.6250	5.6250
Tooth depth, e/i (mm)	h, e/i	5.7761 / 5.6712	5.7761 / 5.6712
Working depth (mm)	h _w	5.0000	
Addendum (mm)	h _a	2.5000	2.5000
Addendum, e/i (mm)	h _a , e/i	2.5000 / 2.4500	2.5000 / 2.4500
Dedendum (mm)	h _f	3.1250	3.1250
Dedendum, e/i (mm)	h _f , e/i	3.2212 / 3.2761	3.2212 / 3.2761
Number of virtual gear teeth	z _n	23.7353	52.2176
Tooth thickness coeff.	s _{nc}	1.5708	1.5708
Transverse tooth thickness (mm)	s _t	4.1790	4.1790
Normal tooth thickness (mm)	s _n	3.9270	3.9270
Normal tooth thickness at d _a (mm)	s _{an}	1.7920	1.9478
Normal tooth thickness at d _a , e/i (mm)	s _{an} , e/i	1.7705 / 1.6730	1.9205 / 1.8336
Normal tooth thickness at d _{Fa} (mm)	s _{Fan}	1.8998	2.0381
Normal tooth thickness at d _{Fa} , e/i (mm)	s _{Fan} , e/i	2.0055 / 1.8973	2.1099 / 2.0164
Normal space width at d _f (mm)	e _{fn}	0.0000	2.0587
Normal space width at d _f , e/i (mm)	e _{fn} , e/i	0.0000 / 0.0000	2.0912 / 2.1107

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	7.8540	
Transverse pitch (mm)	p_t	8.3580	
Normal base pitch (mm)	p_bn	7.3803	
Transverse base pitch (mm)	p_bt	7.7938	
Axial pitch (mm)	p_x	22.9635	
Lead (mm)	p_z	459.2701	1010.3943

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	11.3625	
Length of path of contact, e/i (mm)	g_α, e/i	10.9486 / 10.6327	
Length of approach POC of pinion (A to C) (mm)	g_f1	5.9451	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	5.7336 / 5.6917	
Length of recess POC of pinion (C to E) (mm)	g_a1	5.4174	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	5.2150 / 5.1960	
Length T1 - A (mm)	T1A	3.6639	
T1A tolerance (mm)	e/m/i	4.0601 / 3.9630 / 3.8659	
Length T1 - B (mm)	T1B	7.2327	
T1B tolerance (mm)	e/m/i	7.0207 / 6.9598 / 6.8990	
Length T1 - C (mm)	T1C	9.6091	
T1C tolerance (mm)	e/m/i	9.6186 / 9.6091 / 9.5995	
Length T1 - D (mm)	T1D	11.4578	
T1D tolerance (mm)	e/m/i	11.8539 / 11.7568 / 11.6597	
Length T1 - E (mm)	T1E	15.0265	
T1E tolerance (mm)	e/m/i	14.8145 / 14.7537 / 14.6928	
Length T2 - A (mm)	T2A	27.0850	
T2A tolerance (mm)	e/m/i	26.8526 / 26.7860 / 26.7193	
Length T2 - B (mm)	T2B	23.5163	
T2B tolerance (mm)	e/m/i	23.8804 / 23.7891 / 23.6978	
Length T2 - C (mm)	T2C	21.1399	
T2C tolerance (mm)	e/m/i	21.1609 / 21.1399 / 21.1190	
Length T2 - D (mm)	T2D	19.2912	
T2D tolerance (mm)	e/m/i	19.0588 / 18.9921 / 18.9255	
Length T2 - E (mm)	T2E	15.7225	
T2E tolerance (mm)	e/m/i	16.0866 / 15.9953 / 15.9040	
Length T1 - T2 (mm)	T1T2	30.7490	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	30.7794 / 30.7490 / 30.7185	
Diameter of single contact point A (mm)	d_A	50.1552	121.8596
Diameter of single contact point B (mm)	d_B	51.6826	118.8588
Diameter of single contact point C (mm)	d_C	53.2089	117.0596
Diameter of single contact point D (mm)	d_D	54.6532	115.7755
Diameter of single contact point E (mm)	d_E	58.0089	113.5964
Transverse contact ratio	ϵ_{α}	1.4579	
Overlap ratio	ϵ_{β}	2.0032	
Total contact ratio	ϵ_{γ}	3.4611	
Total contact ratio with tolerance	e/m/i	3.408 / 3.388 / 3.367	
Specific sliding at Tip	ζ_a	0.5244	0.7024
Specific sliding at Root	ζ_f	-2.3601	-1.1026

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	15.0000	19.0000
Profile form deviation (μm)	f_fa	9.0000	9.0000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	10.0000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	28.0000
Runout (μm)	F_r	22.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	33.5000	36.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	3.0000	6.0000
Base tangent length (mm)	W_k	19.2868	42.4310
Base tangent length with tolerance (mm)	W_k, e/i	19.2211 / 19.1835	42.3653 / 42.3277
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	52.8440	116.2899
Theoretical diameter of ball/pin (mm)	D_M.theory	4.2846	4.2249
Effective Diameter of ball/pin (mm)	D_M.eff	4.5000	4.2500
Radial single ball measurement (mm)	M_rk	29.9130	61.4396
Radial single ball measurement with tolerance (mm)	M_rk, e/i	29.8340 / 29.7885	61.3505 / 61.2992
Diametral two ball measurement (mm)	M_dk	59.8260	122.8791
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	59.6680 / 59.5769	122.7010 / 122.5985
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1579 / -0.2491	-0.1782 / -0.2806

3. Detail rating

3.1. General

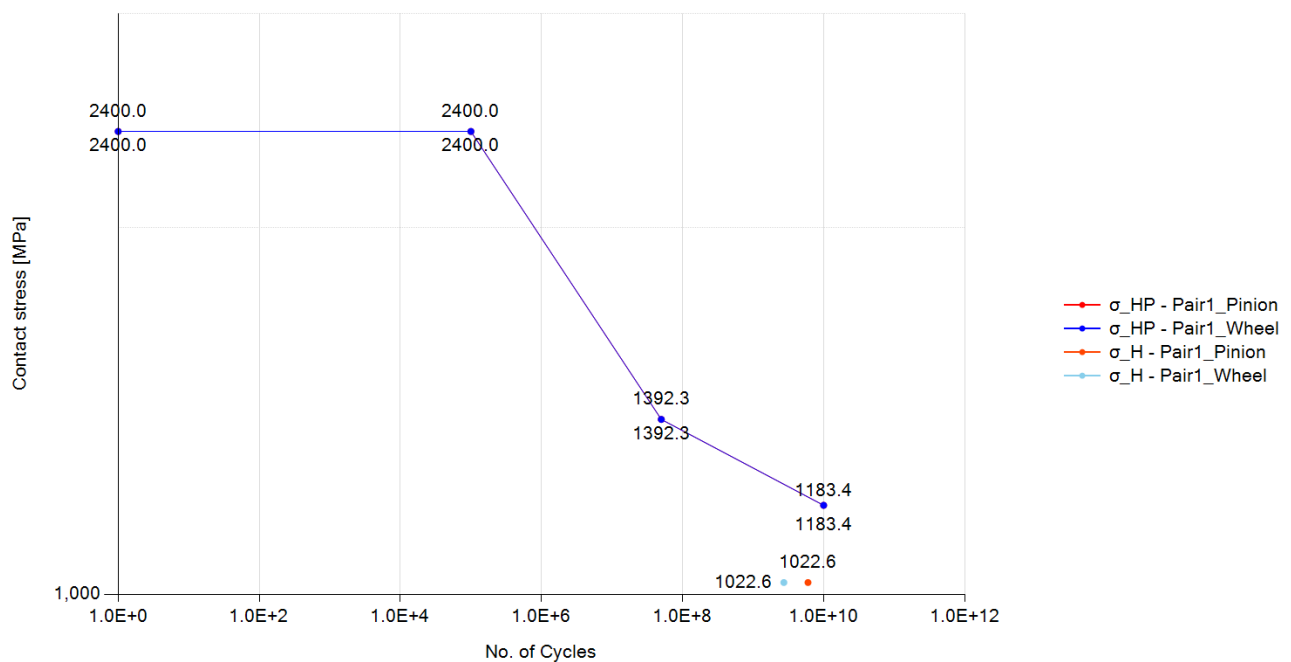
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4545.4550
Nominal tangential load at d (N)	F_t	5384.0420	
Nominal axial load at d (N)	F_a	1959.6310	
Nominal radial load at d (N)	F_r	2085.3960	
Nominal normal load at d (N)	F_n	6097.2900	
Nominal tangential load per unit facewidth (N/mm)	w	117.0440	
Circumferential speed at d (m/s)	v	27.8600	
Sliding velocity at Tip dia. (m/s)	v_g	8.2520	9.0560
Nominal tangential load at d_w (N)	F_{tw}	5384.0420	
Nominal axial load at d_w (N)	F_{aw}	1959.6310	
Nominal radial load at d_w (N)	F_{rw}	2085.3960	
Circumferential speed at d_w (m/s)	v_w	27.8600	
Running-in value (um)	y_p	0.5930	
Running-in value (um)	y_f	0.6750	
Effective single pitch & profile deviation (um)	f_{pb_eff}	7.3080	
Effective single pitch & profile deviation (um)	f_{fa_eff}	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/um)	c'_{th}	17.0300	
Single tooth stiffness (N/mm/um)	c'	11.2090	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/um)	$c_{y\alpha}$	15.0580	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/um)	$c_{y\beta}$	12.8000	
Mean diameter for K_v (mm)	$d_m (K_V)$	52.5840	116.4350
Moment of inertia for K_v (kg.mm ²)	$J (K_V)$	262.1310	4581.8310
Moment of gear inertia per unit facewidth (kg.mm)	J^*	5.6980	99.6050
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0093	0.0334
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0073	
Resonance running speed (rpm)	n_{E1}	21758.4740	
Resonance ratio	N	0.4600	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.1580	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2590	
Transverse load factor - flank	$K_{H\alpha}$	1.1740	
Transverse load factor - root	$K_{F\alpha}$	1.1740	
Number of load cycles	N_L	6.0000×10^9	2.7270×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



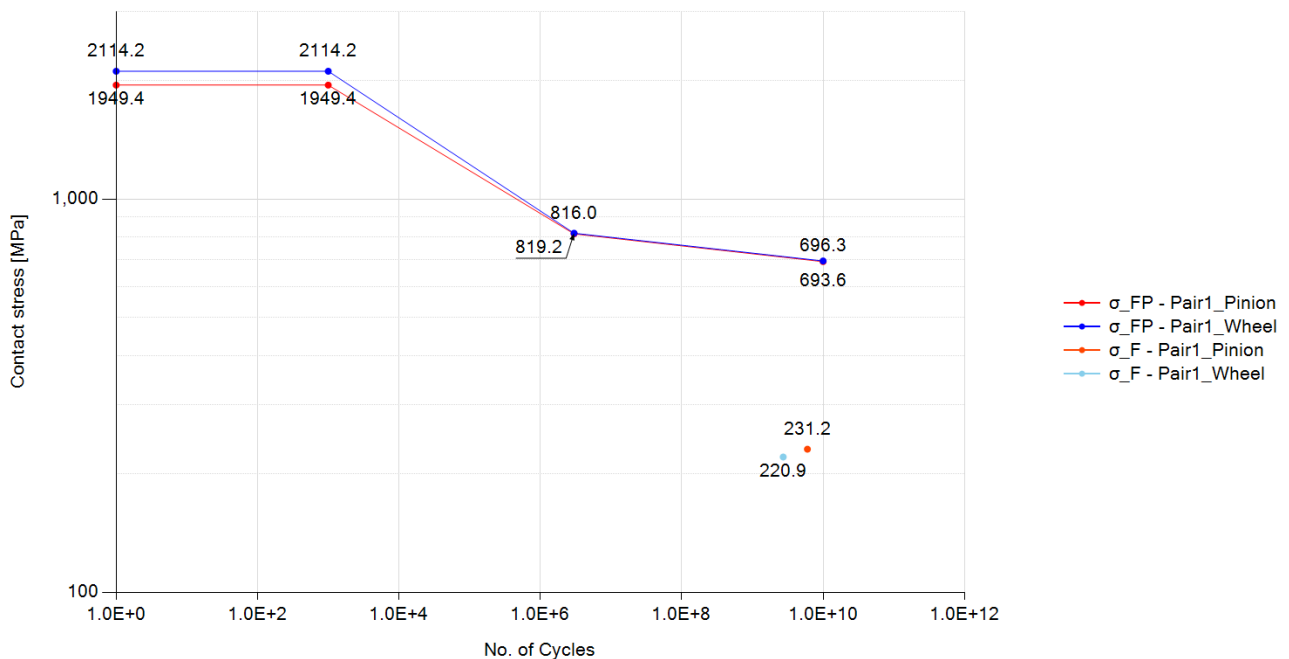
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.8280	
Zone factor	Z_H	2.3710	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0320	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_{H0}	687.8680	
Contact stress (N/mm^2)	σ_H	1022.6200	1022.6200
Life factor	Z_{NT}	0.8630	0.8850

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0300	1.0300
Roughness factor	Z_R	0.9530	0.9530
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1392.2760	1392.2760
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1183.4350	1183.4350
Permissible contact stress (N/mm ²)	σ_{HP}	1202.1240	1231.5520
Safety factor for Contact	S_H	1.1760	1.2040
Contact damage (%)	-	0.5130	0.2330
Cycle to failure	-	1.1690×10 ¹²	1.1690×10 ¹²

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2850	1.1190
Stress correction factor	Y_S	1.7880	1.9620

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.5930	19.8900
Bending moment arm (mm)	h_{Fe}	2.6240	2.7730
Tooth thickness at root (mm)	s_{Fn}	4.9220	5.4020
Tooth root radius (mm)	ρ_F	1.4620	1.3170
Helix angle factor	Y_β	1.0040	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	108.0420	103.2250
Tooth root stress (N/mm ²)	σ_F	231.2210	220.9120
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9920	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	1949.4150	2114.2270
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	815.9540	819.1620
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	693.5600	696.2870
Permissible tooth root stress (N/mm ²)	σ_{FP}	700.6950	714.6500
Safety factor for Bending	S_F	3.0300	3.2350
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	3.2920×10^{26}	3.9020×10^{27}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	39.1000	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	258.6840	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	9.7160	
Angle factor	X_αβ	0.9690	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0470	
Maximum flash temperature (deg)	θ_fl_max	109.6700	
Mean flash temperature (deg)	θ_fl_mean	34.7700	
Total mass temperature (deg)	θ_M	99.6100	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	209.2800	
Safety factor for flash temperature	S_B	2.2030	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0855
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.1426
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.6755
transverse tangential load at reference cylinder per mesh (N)	F_t	5384.0420
transverse load in plane of action (N)	F_bt	5773.8020
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1910
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_By	1.3000
load losses factor	H_v	0.1460
mean coefficient of friction	μ_m	0.0570
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8920
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	16.7290
sliding velocity at point A (m/s)	v_g,A	-9.0560
tangential velocity on pinion at point A (m/s)	v_r1,A	3.8370
tangential velocity on wheel at point A (m/s)	v_r2,A	12.8920
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	102.5500
flash temperature at point A (°C)	θ_fl,A	135.1660
contact temperature at point A (°C)	θ_B,A	237.7150
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3150×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	6.4590×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	v_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	v_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	v_θB,A	1.3690
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	3.4080
material parameter	G_M	2976.0600
local velocity parameter at point A	U_A	7.4090×10 ⁻¹¹
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.0720
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	1087.7350
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1617.0820

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.2000
Tangential speed at working pitch dia. (m/s)	v_{wt}	27.8601
Tangential force at pitch dia. (N)	F_t	5384.0421
Tooth normal force, transverse section (N)	F_{bt}	5773.8016
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	20.1252
Equivalent radius of curvature (mm)	ρ	6.9764
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0374
Tooth loss factor	H_v	0.1465
Mesh power loss (W)	P_{VZP}	821.8890
Mesh efficiency (%)	η_z	99.4521

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4545.4545
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	26.6044	58.5298
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	86241.4604	86241.4604
Froude number	Fr	2974.0100	1351.8227
Centrifugal acceleration (m/s ²)	γ	15919.9334	4277.9606

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	58.2089	122.0596
Pitch diameter of gear (mm)	d_p	53.2089	117.0596
Facewidth (mm)	b	46.0000	46.0000
Dimensionless drag torque	C_m	0.0014	0.0033
Immersion area of the gear (m^2)	S_m	0.0119	0.0321
Total oil churning loss (W)	P	159.2838	1004.8034
Total oil churning torque (N.m)	T	0.1521	2.1109

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000